

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: I-131			FORMS: INORGANIC OR FREE IODINE		
PHYSICAL CHARACTERISTICS HALF-LIFE: 8.02 days DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
364	82	192	90		
637	7	97	7		
284	6	46*	4		
80	3	69	2		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY C – level (low hazard): $\leq 200 \mu\text{Ci}$ B – level (moderate hazard): $> 200 \mu\text{Ci}, \leq 10 \text{ mCi}$ A – level (high hazard): $> 10 \text{ mCi}$					
EXTERNAL RADIATION HAZARDS Gamma dose rate, point source at 1 ft, 1 mCi: 2.2 mrem/h Beta dose rate to skin, point source at 1 ft, 1 mCi: 320 mrem/h Contamination skin dose, uniform deposit of 1 μCi per cm^2 : 6000 mrem/h			INTERNAL RADIATION HAZARDS** Annual Limit on Intake: 30 μCi (Ingestion) 50 μCi (Inhalation) Critical organ: Thyroid The values above indicate the activity taken into the body that would result in 50 rem to the thyroid (CDE). Work with this radioisotope should be done in a proper fume hood.		
SHIELDING Gammas/X-rays: 1 cm of lead will reduce the gamma dose rate by 90%. Betas/electrons: 1.6 mm of plastic will absorb all emissions. Bremsstrahlung may be created and require additional shielding.			DOSIMETRY AND BIOASSAY REQs Whole-body and finger-ring dosimeters are required for handling 5 mCi or more, or 1 mCi amounts weekly . Urine assays may be required after large spills or contaminations. If working with 5 mCi or more, or 1 mCi amounts weekly : Arrange for a thyroid survey within 24-48 hours after the first procedure, and thereafter once every three months.		
SPECIAL PROBLEMS AND PRECAUTIONS: 1. Recommended survey probe: PGM or NaI 2. Volatile iodine solutions should be worked with in a proper fume hood. 3. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often. 4. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required. 5. Segregate waste to those with half-lives of between 8 and 15 days. Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot. 6. Limit soluble waste to the sewer to less than 1 μCi/day per lab.					

- References:
- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2nd ed.). Ashford, Kent: Nuclear Technology Publishing.
 - Johnson, T.E., Birkly, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
 - ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
 - Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
 - Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
 - 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>